9.7 Equal Sums of Powers

WORKSHEET

The question

Let $p_0(x) = (1-x)(1-x^2)(1-x^4)$. Show that $(1-x)^3$ is a factor of $p_0(x)$. If $p_1(x) = xp_0'(x)$ show, by considering factors of the polynomials involved, that $p_0'(1) = 0$ and $p_1'(1) = 0$.

By writing $p_0(x)$ in the form

$$p_0(x) = c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + c_5x^5 + c_6x^6 + c_7x^7,$$

deduce that

$$1 + 2 + 4 + 7 = 3 + 5 + 6$$
$$1^2 + 2^2 + 4^2 + 7^2 = 3^2 + 5^2 + 6^2.$$

Show that we can write the integers $1, 2, \ldots, 15$ in some order as $a_1, a_2, \ldots, a_{15}$ in such a way that

$$a_1^r + a_2^r + \cdots + a_8^r = a_9^r + a_{10}^r + \cdots + a_{15}^r$$

for $r = 1, 2, 3$.

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself If a polynomial $\mathrm{p}(x) = \sum c_k x^k$ has $\mathrm{p}(1) = 0$, what does that say about the coefficients? What about $\mathrm{p}'(1) = 0$?

STAGE 1 A triple root

Ask yourself Factorise $1 - x^2$ and $1 - x^4$. How many factors of $1 - x$ appear?

STAGE 2 Derivatives at 1

Ask yourself If $(1 - x)^3$ divides p_0, what power of $(1 - x)$ divides p_0'? And then $\mathrm{p}_1 = x\mathrm{p}_0'$ and p_1'?

STAGE 3 Reading off the coefficients

Ask yourself What are $\mathrm{p}_0'(1)$ and $\mathrm{p}_1'(1)$ in terms of the coefficients c_k?

STAGE 4 One more factor

Pause and try To make the cubes balance too, we need a fourth factor of $(1 - x)$. Which polynomial should replace p_0?

Try this next

1 **Shifting.** (From STEP II, 2025.) Suppose $\sum a_k = \sum b_k$ and $\sum a_k^2 = \sum b_k^2$ for two lists of the same length m. Show that $\sum a_k^3 + \sum (c + b_k)^3 = \sum b_k^3 + \sum (c + a_k)^3$ for every c. Use this, starting from a balanced split of $\{1, \ldots, 8\}$, to find a split of $\{1, \ldots, 16\}$ in which the cubes balance as well.

2 **Squares of squares.** (Also from STEP II, 2025.) The sets $A = \{1, 3, 4, 5, 9, 11\}$ and $B = \{2, 6, 7, 8, 10\}$ have equal sums and equal sums of squares. Split $\{n^2, (n+1)^2, \ldots, (n+11)^2\}$ into two sets with equal sums, for every positive integer n. (*Hint:* add 0 to the second set.)

3 **Translation invariance.** Show that if two sets of numbers have equal sums of rth powers for $r = 0, 1, \ldots, k$, then so do the sets obtained by adding the same number c to every element. Why does this mean that the question's sets $\{1, \ldots, 15\}$ could

equally be replaced by any 16 consecutive integers?

4 Fair sweets. Sweets have values $1, \frac{1}{2}, \frac{1}{3}, \ldots, \frac{1}{8}$. Compare the totals A gets with the orders $ABABABAB$ and $ABBABAAB$.

Further reading

- **Jean-Paul Allouche & Jeffrey Shallit, "The ubiquitous Prouhet–Thue–Morse sequence"**, in *Sequences and Their Applications* (Springer, 1999). A survey of the sequence's many appearances; free online.
- **Matt Parker, *Things to Make and Do in the Fourth Dimension*** (Penguin). Includes equal power sums and other number curiosities.
- **Joshua Cooper & Aaron Dutle, "Greedy Galois games"**, *American Mathematical Monthly* 120 (2013). Fair turn-taking and Thue–Morse.

The full guided solution, and what the question leads to at university, is in Section 9.7:
https://beyondstep.keithmcnulty.org/ch09/equal-power-sums.html